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CO_2 maritime transportation

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ABSTRACT

The objective of this study is to describe the complete transport chain of CO_2 between capture and storage including a ship transport. This last one is composed by the following steps:

Shore terminal including the liquefaction, temporary storage and CO_2 loading.
Ship with a capacity of 30,000 m^3.
On or off shore terminal including an unloading system, temporary storage and export towards the final storage.

Between all the possible thermodynamic states, the liquid one is most relevant two options are compared in the study (−50 °C, 7 bar) and (−30 °C, 15 bar). The ship has an autonomy of 6 days, is able to cover 1000 km with a cargo of 2.5 Mt/year. Several scenarios are studied varying the geographical position of the CO_2 source, the number of harbours and the way the CO_2 is finally stored.

Depending on the option, the transport cost varies from 24 to 32 €/tCO_2. This study confirms the conclusion of a previous study supported by ADEME, the cost transport is not negligible regarding the capture one when ships are considered. Transport by ship becomes a more economical option compared with an off shore pipeline when the distance exceeds 350 km and with an onshore pipeline when it exceeds 1100 km.

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1. Introduction

An answer to climate change is CO_2 capture/transport and storage (CCS). This answer consists in the capture of CO_2 from large point sources emissions and the find of safe place to store it. But source and sink are generally far one from the over. To transport large amount of CO_2, pipeline seems to be the best solution. And a previous study, Bonnissel et al. (2007) designed a supercritical pipeline and gave cost estimation. But transporting CO_2 by ship will be far more flexible and less expensive for long distances. Gas carriers can also be used to collect CO_2 from a number of plants for transport to a central staging post-connected to an aquifer by an off shore pipeline. Four existing ships, transporting carbon dioxide for use in industry and alimentary processes have small capacities: between 800 and 1200 m^3.

Few economical studies have been realised including a ship transport, Svensson et al. (2007), Aspelund et al. (2007). In this last one the complete chain is described including the intermediate storage, loading and unloading systems and the ship. A combined LPG/CO_2 semi-refrigerated ship is chosen with a capacity of 20,000 m^3 under (−52 °C, 6.5 bar). A complete energy and cost estimate analysis is performed concluding that the total cost is 20–30 USD/tCO_2 for volumes larger than 2 Mt/year and distances limited to North Sea.

This paper presents a complete transportation chain study including CO_2 conditioning, pipeline transport, liquefaction, loading/unloading systems, temporary storages and a new 30,000 m^3 ship design. A cost analysis based on three scenarios is also described.

2. Transportation chain description

2.1. Capture plants and conditioning

Three flowrates are studied: 1, 2 and 3 Mt/year (this last one corresponding to a coal steam power plant of

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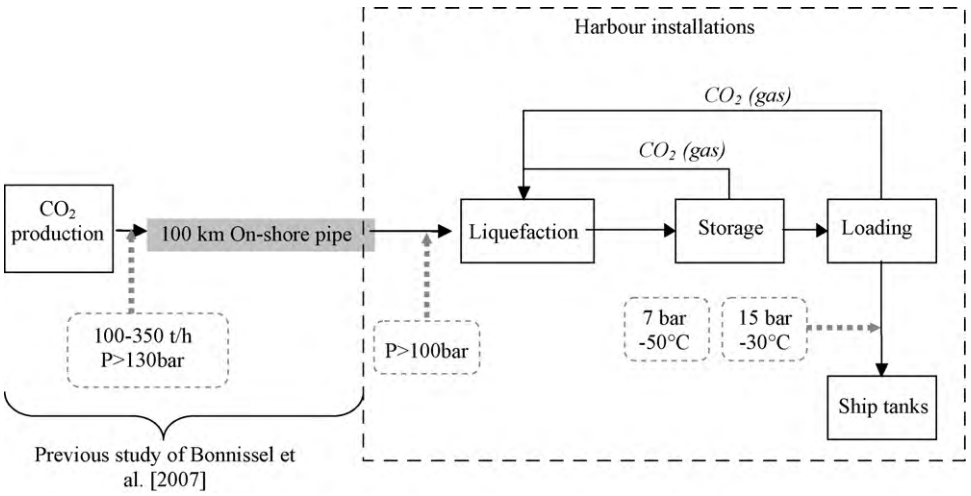


Fig. 1. Basic cases.

600 MWth). For the study, CO_2 is considered free of impurities and water. Two CO_2 sources locations are considered (Fig. 1):

- a source at 100 km fare from the harbour. CO_2 is transported in a supercritical phase by a 15 in. internal diameter pipeline. It is delivered at the terminal at 100 bar and at ambient temperature. The gas is available at 1.7 bar with a content of 28% of water after capture, CO_2 conditioning is presented on Fig. 2 and described in Bonnissel et al. (2007); The evaluation performed in this study leads to a total cost of € 83 million for the compression & dehydration process, and between € 78 million and € 90 million for the pipeline.
- a source in the harbour area. CO_2 is delivered at 1.7 bar; 25 °C, at the liquefaction stage.

2.2. Liquefaction, storage and loading

Between all the possible options of transport we selected two thermodynamic conditions of liquid CO_2: a low temperature to limit tank cost thanks to lower pressure (−50 °C, 7 bar) and a high temperature to limit energy penalty in the liquefaction plant

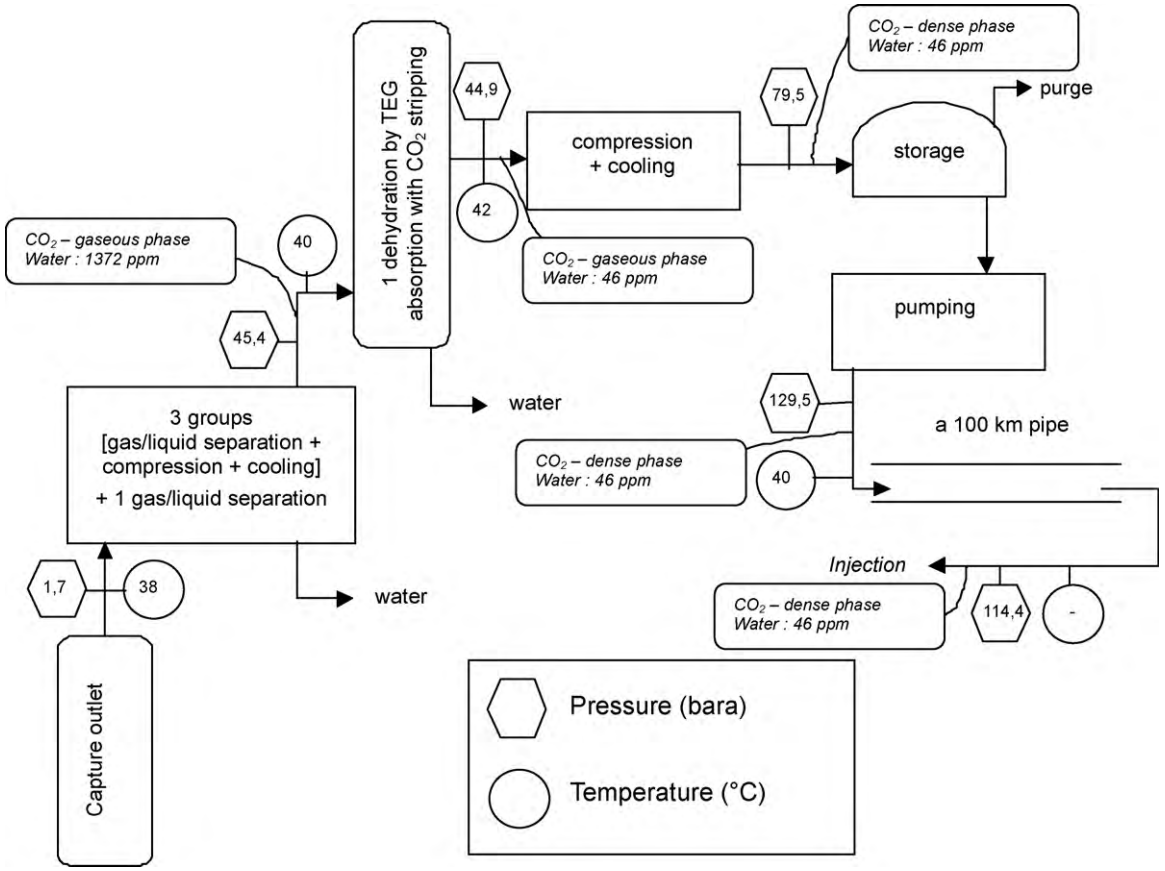


Fig. 2. Process flow diagram for the dense phase case.

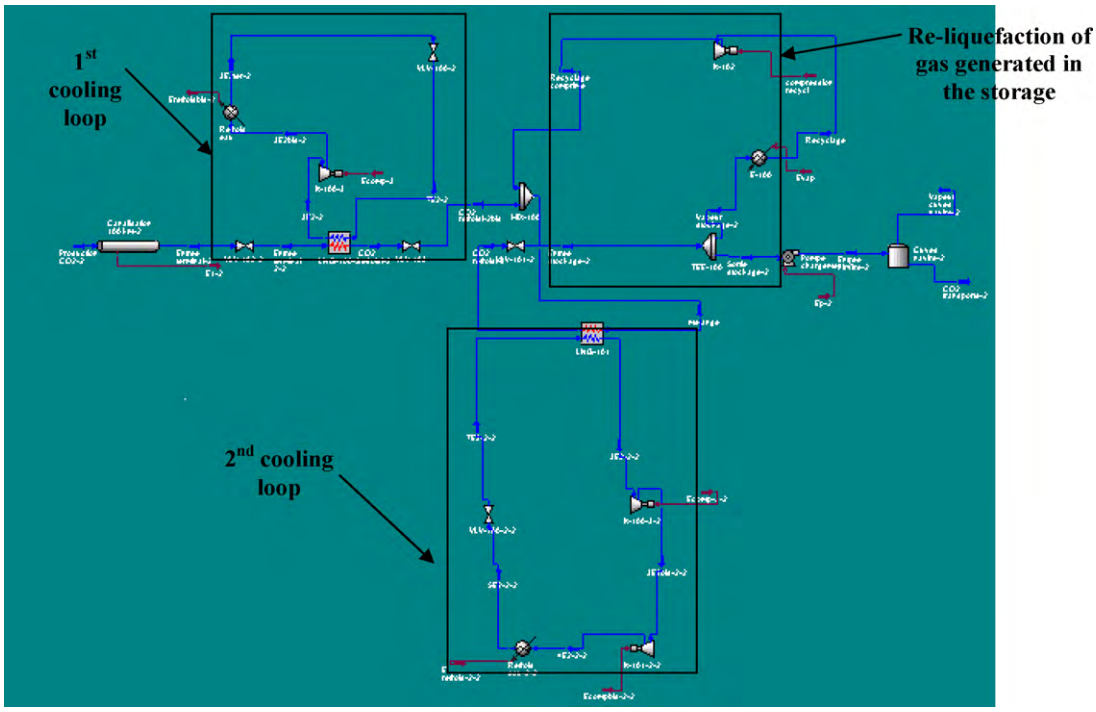


Fig. 3. Liquefaction unit.

(−30 °C, 15 bar). We avoid solid CO_2 transport because of the impact on the loading/unloading operations, solid handling being much more difficult than liquid, especially in case of off shore unloading operation.

2.2.1. The liquefaction

For both sources the liquefaction chain has been simulated using the software HYSYS.

From a bibliographic study and taking into account the new regulations, two coolers are compared in the study: propane and propylene. Two scenarios of cooling are taken into account (Fig. 3):

- CO_2 is first compressed or expanded to reach 7 bar (1 or 2 compressor stages depending on the refrigerant nature) or 15 bar (2 or 3 compressors stages depending on the refrigerant nature)

Table 1
For a source at 100 km from the harbour, energy required from the different options.

	CO_2 flow rate = 350 t/h		-50 °C / 7 bar	-30 °C/ 15 bar	-30 °C/ 15 bar	-30 °C/15 bar
	Fluid cooling loop Nature	(% mass.)	20% ethylene - 80% propylene	20% ethylene - 80% propylene	100% propane	100% propane
	Flow rate	(t/h)	**193**	**216**	**143**	**147**
	Compressor power					
CL1	Stage 1	(MW)	*from 1.4 to 3.7 bar* 4.4	*from 3.1 to 8 bar* 4.6	*from 1.3 to 3.7 bar* 3	*from 3.25 to 9.75 bar* 1.8
CL1	Stage 2	(MW)	*from 3.7 to 9.5 bar* 4.6	*from 8 to 21 bar* 4.7	*from 3.7 to 10 bar* 2.9	
CL1	Stage 3	(MW)	*from 9.5 to 24 bar* 3.9			
CL2	Stage 1	(MW)				*from 1.3 to 3.6 bar* 1.1
CL2	Stage 2	(MW)				*from 3.6 to 9.8 bar* 1.1
	CO_2 Pomp power	(kW)	34	36	36	36
	Energy					
	Energy for liquefaction	(kWh/tCO2)	37	27	17	12
		(kJ/kg)	**133**	**96**	**61**	**42**
	Upstream energy*	(kWh/tCO2)	94	94	94	94
		(kJ/kg)	338	338	338	338
	Total energy	**(kWh/tCO2)**	**131**	**121**	**111**	**106**
		(kJ/kg)	**471**	**434**	**399**	**380**

*Energy for dehydratation and compression before entering the 100 km pipeline.

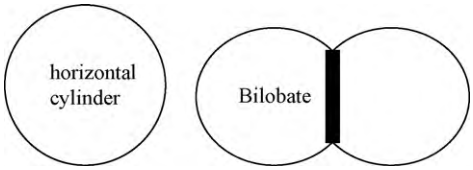


Fig. 4. Possible tanks for CO_2 temporary storage.

then cooled down to −50 °C or −30 °C and expanded again to be liquefied.
- 2 cooling loops in serial. The first compressor stage provides a CO_2 at −7 °C, which is then expanded down to 30 bar. The second loop cooled down the CO_2 to −30 °C and the last expansion is realised to reach 15 bar.

Table 1 summarises the energy required for all these options for a source located 100 km fare from the harbour. It is clear that the options propane, double cooling loops lead to the best scenario. This conclusion is confirmed in case of CO_2 capture in the harbour area.

2.2.2. The temporary storage

To simplify the design and reduce costs, we considered that terminals and ships storage are composed of the same tanks. Storage on terminal is twice the storage on ship. Two designs are possible: cylinder or bilobate (Fig. 4).

The design satisfies the BS5500 PD code. It is realised with the sailing conditions and takes into account the fact that the boil off shall be neither released to the atmosphere, nor reliquefied. Then a possible pressure increase from 7 to 10.4 for the option (−50 °C, 7 bar) and from 15 to 19.5 for (−30 °C, 15 bar) and two days of margin in case of bad weather condition are integrated in the design conditions. The design temperature corresponds to the one obtained during a sudden depressurisation and is taken equal to −80 °C. This critical temperature governs the choice of the quality of the steel: 3.5%, 5% and 9% Ni; stainless steel 304L and 316L; aluminium 1050. The use of stainless steels and aluminium leads to high thickness and possible construction difficulties. 5 and 9% Ni are the best candidates, 5% Ni being the cost optimum solution and 9% Ni being the weight optimum solution. We chose the 9% Ni which leads to smaller supplying delays and is well known by the manufacturers. The optimal solution is a cylindrical tank in steel 9% Ni, with a casing of 10 mm thick, full of perlite (Fig. 5).

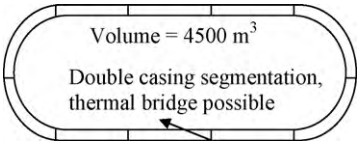


Fig. 5. Intermediary storage characteristics. A 4500 m^3 tank composed of a double casing containing Perlite vacuum packed (thickness 30 cm), having an external diameter of 14.7 m for 32.3 m length.

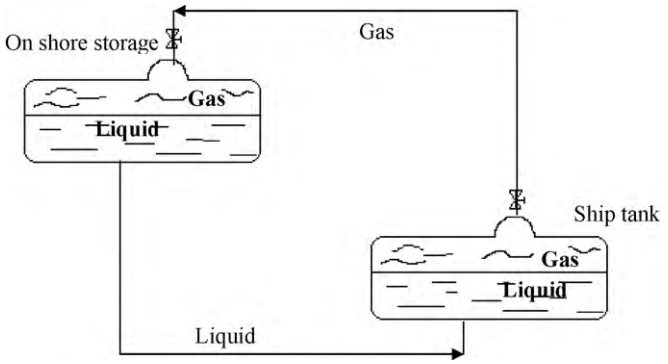


Fig. 6. Boil off management during the loading process.

2.2.3. The loading process

In order to transfer the liquefied CO_2 from the temporary storage to the ship, a system of pumps and flexible or rigid arm (piping) is designed. These lines have a diameter of 16 in. and a length of 120 m. Pressures need to be equilibrated between both storages as the boil off (CO_2 evaporation) generated during the loading operation increases the ship tank pressure. The gas generated is sent back to the liquefaction unit by a second line (Fig. 6).

2.3. Ship description

Gas is usually transported under its liquid phase, than means under low temperature. As the ship is specified without any reliquefaction plant onboard, tanks are designed to resist to the pressure increase.

Specifications for the ship are:

- A capacity of 30,000 m^3,
- Process for a loading and unloading with an onshore terminal,

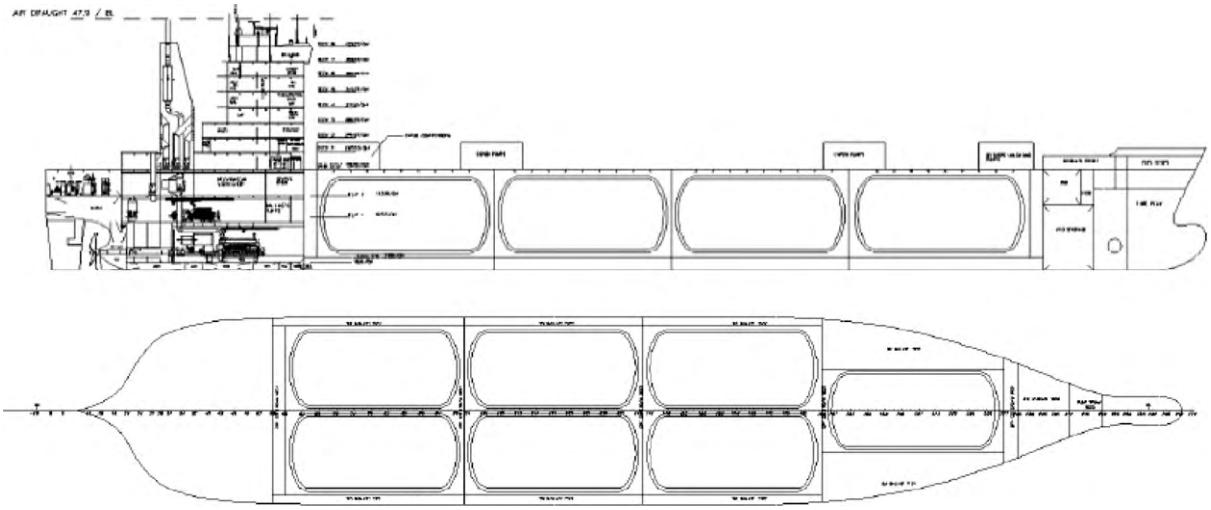


Fig. 7. Typical arrangements for a CO_2 ship.

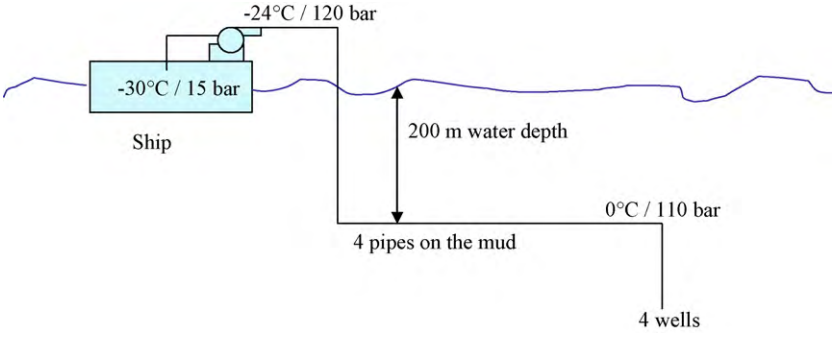


Fig. 8. Offshore unloading.

- Pumps with a total power of 5.2 MW, for an offshore unloading,
- Dynamic position for offshore unloading,
- The most economical propulsion in order to reach a speed of 16.5 knots.
- Following the existing regulations: SOLAS (Safety Of Life At Sea), IGC (Ship Carrying Liquefied Gas in Bulk) and MARPOL (MARitime POLution).

Based on these specifications a ship is designed (Fig. 7). 7 tanks, as the one presented previously, are integrated. Cofferdams (600 mm) are placed between the tanks and the front and back parts of the ship in order to limit thermal exchanges and allow inspections. Possible dilatation of the tanks was considered in the design of a new fixing system including wood parts.

Different solutions for the propulsion are considered:

Architecture	Advantages	Disadvantages
Electric-diesel	Better consumption, architecture optimisation, low power installed	CAPEX high, complexity (higher breakage risk), site coverage consuming
Diesel (type CODAD, COmbinaison Diesel Alternateur Diesel)	CAPEX, well known technology, less breakage risk than electric-diesel	OPEX (higher consumption than electric-diesel)
2T motors	CAPEX, well known technology, OPEX	Pollution, no dynamic position, heavy

The best option is the electric-diesel but could be considered as too much expensive by the future clients for sequestration operations. Then a more classical architecture like the CODAD type (diesel) is our final choice. Two motors, fed by heavy fuel oil, will be connected to an adjustable blade propeller (Diameter 7 m). The energy necessary on board will be provided by four diesel alternators.

Installation of the different utilities on board is realised in parallel with the stability analysis of the ship. The stability analysis is realised using the software ARGOS with a working load from 10% to 100% (tank full at 95%, keeping a volume for the boil off). It takes into account the new regulations on biologic prevention pollution (ballast are full of water, their loading procedures are described) and the IGC code relative to possible damages. As the ship as an axial symmetry, all the damages are simulated at port on a watertight bulkhead. ARGOS simulations show that IGC criteria are completely respected.

2.4. On or offshore unloading

2.4.1. Onshore terminal

This terminal is similar to the loading one. Temporary storage will be twice the one present on the ship in order to have a safety margin, this hypothesis could be optimised integrating the management of a complete fleet of CO_2 ships. Possible reuse of the liquid CO_2 frigories will have to be integrated in a more detailed study.

2.4.2. Offshore unloading

Between all the possibilities a direct unloading is considered. In order to fit with the constraints: maximum time on place 24 h in the North Sea, injection power of 5.2 MW for 30,000 m^3 of CO_2 at −30 °C, 15 bar (Fig. 8). To be able to respected them, 4 wells are necessary to achieve a flowrate of 350 t/h, maximum value tolerated by the aquifer. Conditions imposed at the well head are 0 °C and 110 bar. Pumps on the ship will furnish the energy to reach 95 bar, the static pressure over the 200 m water depth, will complete it. During this phase, CO_2 temperature increases a bit. To avoid more energy consumption and CO_2 release, CO_2 is naturally heated by thermal exchange between a Bare pipe and sea water, before reaching the well head. To increase thermal exchanges the pipe (X60) is not buried and is only coated against the corrosion. Due to the very low temperature at the inlet of the pipe, some ice will form on external pipe wall over few kilometres. Table 2 summarises different possibilities depending on currents and internal pipe diameter. The optimum between steel mass and pressure drop leads to the choice of a 10 in. NPS (internal diameter of 10 in.) for a designing pres-

Table 2
Pipe length necessary to reach the injection conditions.

	Solution with 4 injection lines							
Flow rate per well (bpd)	49,528	49,528	49,528	49,528	49,528	49,528	49,528	49,528
External current velocity (m/s)	0.005	0.5	0.005	0.5	0.005	0.5	0.005	0.5
Pump outlet temperature (°C)	−22	−23	−23.7	−24.2	−24.2	−24.4	−24.4	−24.5
Internal diameter (")	8	8	10	10	12	12	14	14
Pressure drop (bar)	21.0	9.0	6.2	2.4	2.2	0.7	0.9	0.3
Subsea pipe length (km)	7	3	6.5	2.5	6	2	5.5	2
Ice deposition length (km)	3		3		3		2.5	
Injection temperature (°C)	0.1	−0.4	0.0	−0.6	0.0	−1.6	−0.1	−0.3
CO_2 velocity (m/s)	2.8	2.8	1.8	1.8	1.2	1.2	0.9	0.9
Steel mass (t)	671	288	974	375	1295	432	1616	587
Pipe volume (m3)	908.0	389.2	1317.4	506.7	1751.2	583.7	2184.9	794.5
Topside pump power (MW)	5.43	5.55	5.18	4.76	4.73	4.58	4.59	4.53

sure of 180 bar. Assuming the minimal external temperature of 6 °C (winter North sea), less than ten kilometres are necessary to reach 0 °C. Weaknesses of this solution are the embrittlement of the pipe due to freeze/thaw cycles, the sensibility to currents, regulations obliging the burrial of the pipes. Simulations, adding a temporary storage (on plateforme or subsea) have been realised. Such solutions are possible, but with weak points such as the necessity of an high floor space, ballasts and subsea structure appear. Other kind of solutions could be used depending on the existing installations: use of a jacket for mooring, CALM systems (Catenary Anchor-Leg Mooring), etc. CALM buoys are frequently used in offshore industry for the oil offloading of floating production facilities. In TransCO_2 case, CO_2 carrier is moored to the CALM buoy the liquid CO_2 is transferred from the carrier to the injection network with floating hose. For this solution there is a need for development concerning low temperature and high pressure flexible. A solution without mooring structure can be envisaged but required a dynamic positioned (DP) vessel. This kind of vessel is equipped with thrusters ensuring the perfect positioning of the vessel over the injection location. DP vessels are commonly used for the installation of subsea pipelines and drilling rig, but they are fuel consuming degrading the carbon balance of the CO2 transport. These solutions are compared in the following table:

Mooring type	Advantage	Weakness
No mooring structure	Vessel with dynamic positioning	Fuel consuming CO_2 emission Weather dependant
CALM system	Control of ship position Weather independent Well known technology	Mooring assistance Space consuming
Disconnectable mooring plug	Control of ship position Weather independent Self-governing mooring	
Jacket	Control of ship position Weather independent Reuse of existing structure	Costly if new Mooring assistance

3. Scenarios

The basic harbour is composed of

- liquefaction part: centrifugal compressors, plate heat exchanger, vertical centrifugal pump (high flow rate, axial flux), cooler boxes and additional pumps.
- storage: 14 tanks, lines between storage (on terminal or on ship) and liquefaction unit (boil off treatment), flexible or rigid arm, transfer pump.

Depending on the scenario, functionalities on the harbour will be adjusted.

Three scenarios are compared (Fig. 9):

1. 2 harbours A and B, B having a direct connection by pipe (200 km) to an off shore aquifer. A liquefaction unit is present on B.
2. 2 harbours A and B, there is no liquefaction unit on B. Then ambient temperature of the CO_2 coming from the on shore pipe and directly connected to the off shore pipe is used to heat the CO_2 coming from the ship.
3. 3 harbours A, B and C, A and B correspond to the basic case, C is connected to an off shore aquifer by a pipe and doesn't have a liquefaction unit. These last case could be representative of a French, English and Dutch case.

4. Cost estimate

An economical study is performed on the scenarios previously described. For this estimation the complete transport chain is considered: conditioning of the CO_2 at the outlet of the capture unit (only dehydratation and compression), 100 km onshore pipe in supercritical conditions (180 bar), liquefaction unit, temporary storage, ship transport covering 1000 km, unloading in an offshore pipe for final storage. Economic depreciation is supposed to be 5% over 30 years. Costs for the conditioning and the on shore

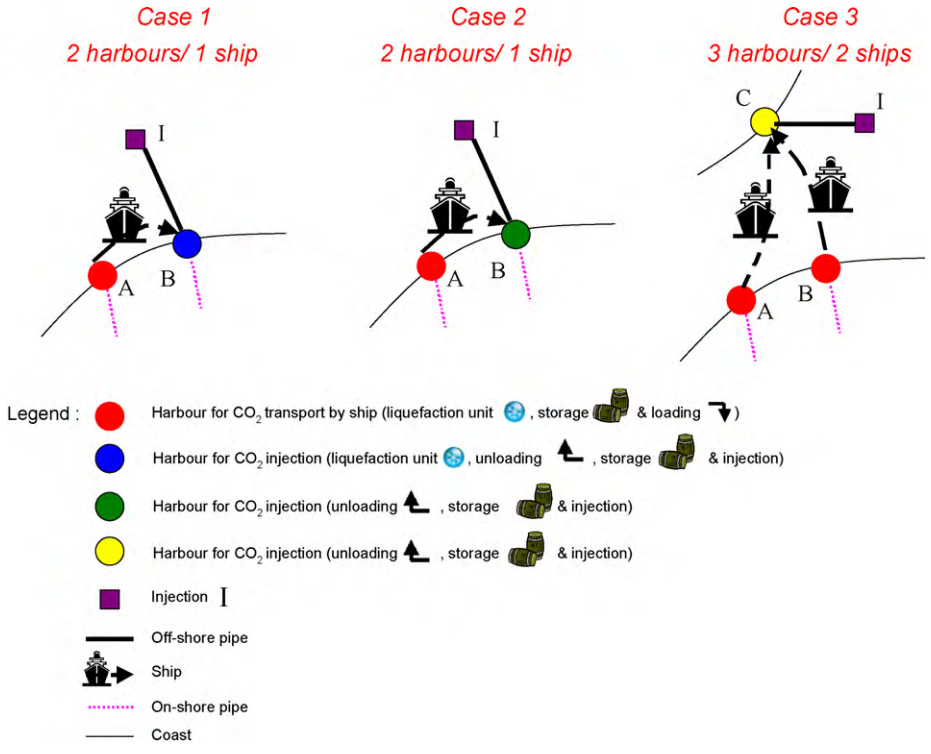


Fig. 9. Use cases.

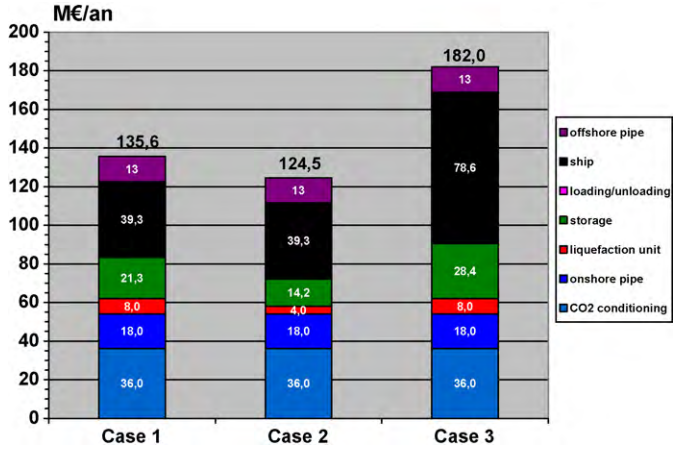


Fig. 10. Scenarios costs evaluation.

transport are based on a previous study supported by ADEME (Bonnissel et al., 2007). and based on Chauvel cotations (Chauvel et al., 2003).

For both solution (−50 °C, 7 bar) and (−30 °C, 15 bar), the total cost (CAPEX and OPEX) of the installations from the outlet of the capture plant to the inlet of the temporary storage (conditioning, transport and liquefaction) showed that the (−30 °C, 15 bar) case was the cheapest one. Then scenarios were estimated only for this option.

For the liquefaction the case (−30 °C, 15 bar) and 2 cooling loops is considered.

Transport chain	Description	Cost M€/year (economic depreciation 5% over 30 years)	Source (year price)
CO2 conditioning	Dehydratation, compression	18	2005
On shore transport	Onshore pipe	9	2005
Liquefaction unit	Compressors, heat exchangers, pumps, etc.	4	2005
Storage	14 tanks	7.1	2007
Loading/unloading	Pump, loading arm, transfer lines	Negligible	
Ship	Ship with 7 tanks	39.3	2008
Injection	Heating, heat exchangers, pump		
	Subsea flow line	13	2008

Fig. 10 present costs obtained for the three scenarios. Ship transport represents the most expensive part in the chain. However all the expenses linked with a normal ship exploitation are taken into account within this cost i.e. the one of the ship, the expenses for her maintenance, the crew, tax, insurance, harbours, etc. The conditioning part is under estimated as it is composed in our examples of only the dehydratation and the compression units. In reality, impurities will have to be treated too, increasing the final cost of the conditioning part.

Respectively for the scenarios 1, 2 and 3, 24 €/tCO_2, 22 €/tCO_2, and 32 €/tCO_2 are obtained. This confirms that transport cost is not negligible when it integrates a ship part.

A comparison between ship transport and on or off shore pipe is performed. Depending on the CO_2 quantity, more than one ship can be considered. CO_2 source is considered to be in the harbour area and then only the CO_2 conditioning, liquefaction and storage are considered. Cost for the pipe onshore are those considered in the previous study (Bonnissel et al., 2007). Fig. 11 presents the results of this comparison. 4 quantities of CO_2 are simulated (0.8, 1.6, 2.8, 5.6 Mt/year). Discontinuities of the lines are due to the fact that

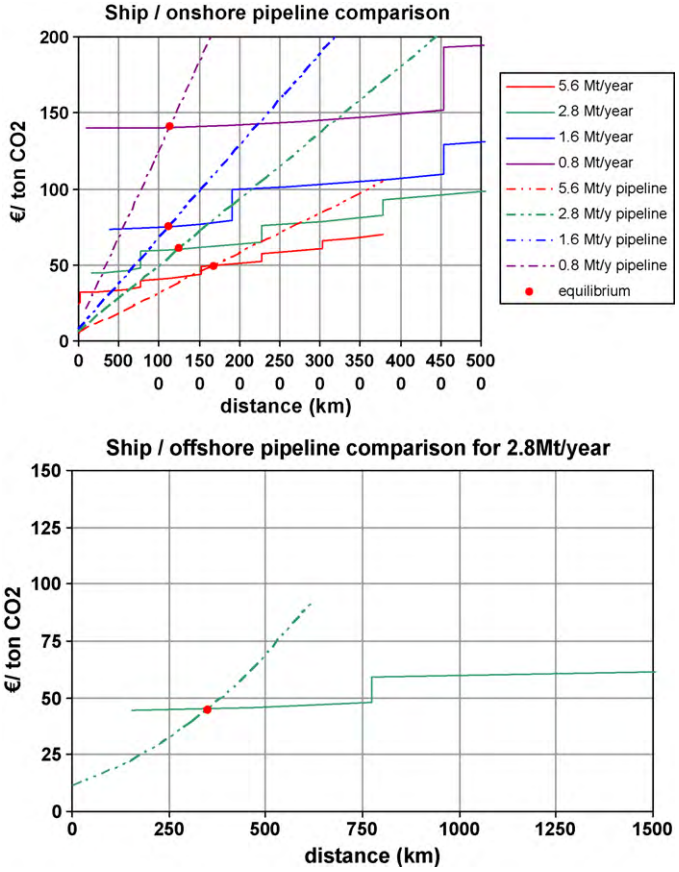


Fig. 11. Ship/pipeline comparison.

only a whole number of ships can be added. Clearly our 30,000 m^3 ship is over estimated for the 0.8 Mt/year and at least two ships are necessary to transport 5.6 Mt/year. Then compared to a pipeline ship is a more economical solution when distance between two harbours exceed about 1100 km (onshore pipe comparison), and exceed 350 km compared to an offshore pipeline (export towards an offshore aquifer).

5. Conclusion

Ship transport is a flexible alternative to pipeline for an offshore storage option, even for a harbour to harbour transport. It can be cost attractive for long distances (350 km for an offshore storage, 1100 km for a coast to coast case). Compared with the pipe option. This technical-economics study shows the feasibility to build and use a 30,000 m^3 ship to transport CO_2 under (−30 °C, 15 bar). Cost estimation leads to a transport cost from 24 to 32 €/tCO_2 covering the chain from the capture outlet to the injection in an offshore aquifer.

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References

Aspelund, A., Mølnvik, M.J., De Koeijer, G., 2007. Ship transport of CO_2 technical solutions and analysis of costs, energy utilization, exergy efficiency and CO_2 emissions. Chemical Engineering Research and Design 84 (A9), 847–855.

Bonnissel, M., Broutin, P., Fradet, A., Odru, P., Ruer, J., Saysset, S., 2007. Technical and economic assessment of CO_2 transportation for CCS purposes. The Journal of Pipeline Engineering 6 (3), 173–179.

Chauvel, A., Fournier, G., Raimbault, C., (new, revised and expanded edition) 2003. Manual of Process Economic Evaluation. Éditions Technip, Paris.

Svensson, R., Odenberger, M., Johnsson, F., Strömberg, L., 2007. Transportation Infrastructure for CCS – Experiences and Expected Development. Chalmers University of Technology and Vattenfall, Sweden.